Relatedness and ancestry inference from pangenome-derived alignments

IBS, IBD, and local ancestry without variant calling

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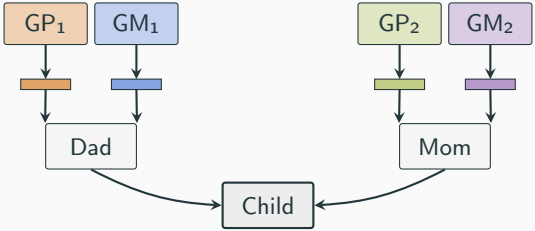
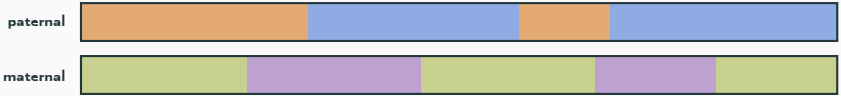
A chromosome inherited through a pedigree

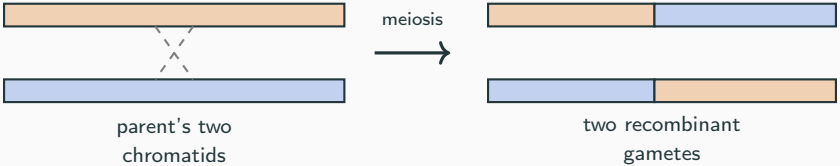


child's chromosome 1 (both homologs):

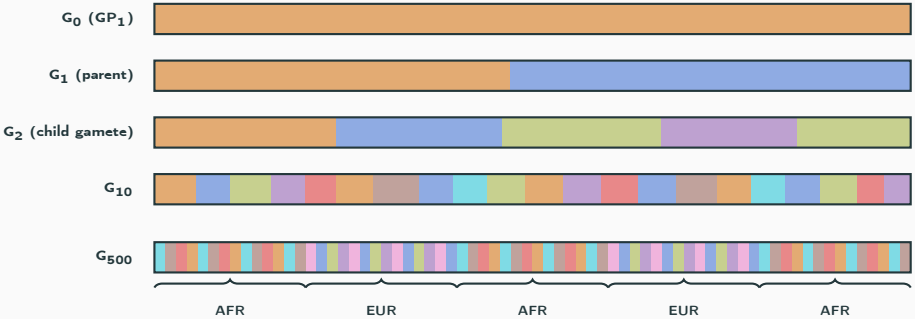


Through-generation recombinations produce mosaics.

Recombination at meiosis



Segment lengths shrink across generations



At G_{500}, too many people contribute for individual ancestors to matter; what stays meaningful is the population label of each region.

Pedigree and ancestry: two time scales

pedigree perspective (recent: 1–50 generations)

ancestry perspective (deep: 100s–1000s generations)

Two inference questions: kinship and ancestry

Kinship (IBD)

did this segment come from a *recent* common ancestor?

Ancestry (LAI)

which *population* is this segment most consistent with?

Kinship: identity from a recent common ancestor

expected de novo mutations in 1 Mb over 5 generations:

$$\mu \cdot t \cdot L \;=\; (1.2\times10^{-8})(5)(10^{6}) \;\approx\; 0.06.$$

Essentially identical. Detection target: identity ≥ 0.999 across many adjacent windows.

Ancestry: similarity, not identity

We compare a query haplotype to a panel that *represents* a population — not the actual ancestor.
Two things make exact identity unrealistic:

1. Panel diversity

panel haplotypes differ from each other and from the true ancestor

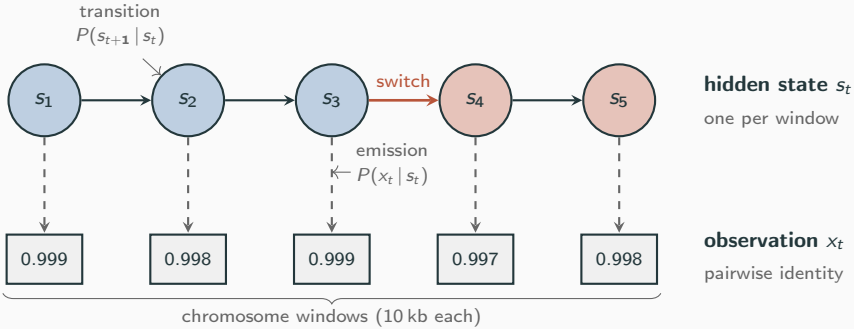
2. Mutation accumulates

100s–1000s of generations of substitutions on both query and panel lineages

Current methods for IBD and ancestry inference

family	examples	painting layer	task
HMM (Li–Stephens)	RFMix, FLARE, ChromoPainter	forward–backward	ancestry
Machine learning	Gnomix	ML + smoothing	ancestry
ARG-based	AncestralPaths	NN on genealogical neighbours	ancestry
Graph transformer	ARGMix (Shanks 2026)	self-attention on coalescent trees	ancestry
HMM (IBD)	IBDseq, BEAGLE-IBD, PARENTE2	forward–backward	IBD
IBD seed-and-extend	hap-ibd, iLASH, GERMLINE	local match extension	IBD

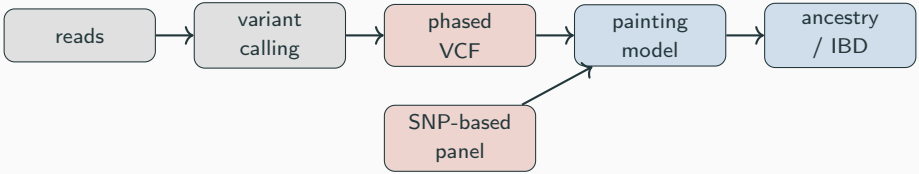
The hidden Markov model behind every part



Part 2 (IBD)	state alphabet {IBD, not-IBD}
Part 3 (LAI)	state alphabet {AFR, EUR}
Part 4 (pedigree)	state alphabet {GP_PAT, GM_PAT, GP_MAT, GM_MAT}

Same skeleton, different alphabet. Transitions calibrated by recombination (`-p-enter-ibd`); emissions learned (`-baum-welch-iters`) or parameterised (`-auto-configure`).

Shared upstream pipeline

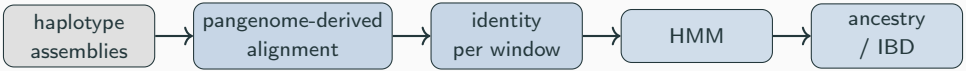


the variant catalog

Costs of the variant catalog

- SNP ascertainment bias
- Structural variants invisible
- Phasing errors at ancestry boundaries
- Reference-panel dependency

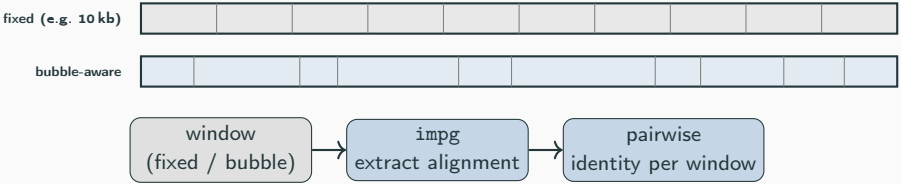
Bypassing the variant catalog



no variant calling

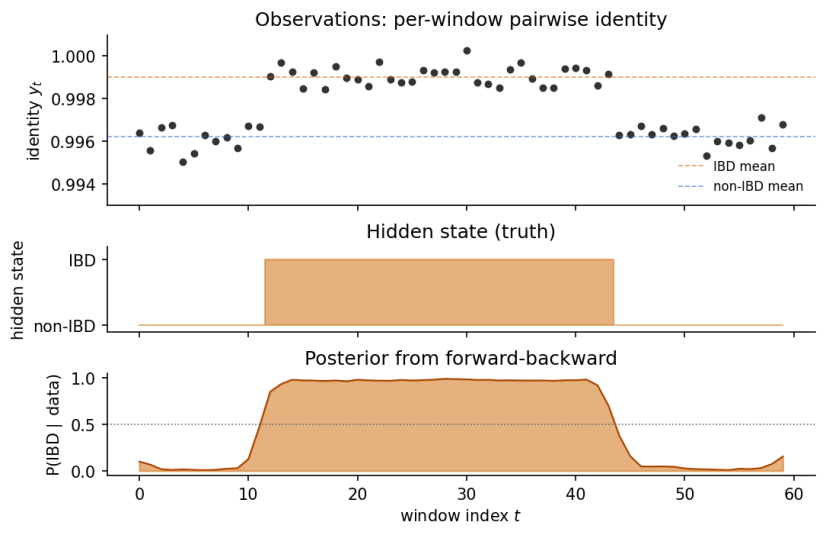
Per-window similarity from pangenome alignments

Two ways to partition the chromosome before querying impg:



Identity is read *directly off the alignment* — no SNP calling, no variant catalog.

Hidden Markov model: per-window state inference

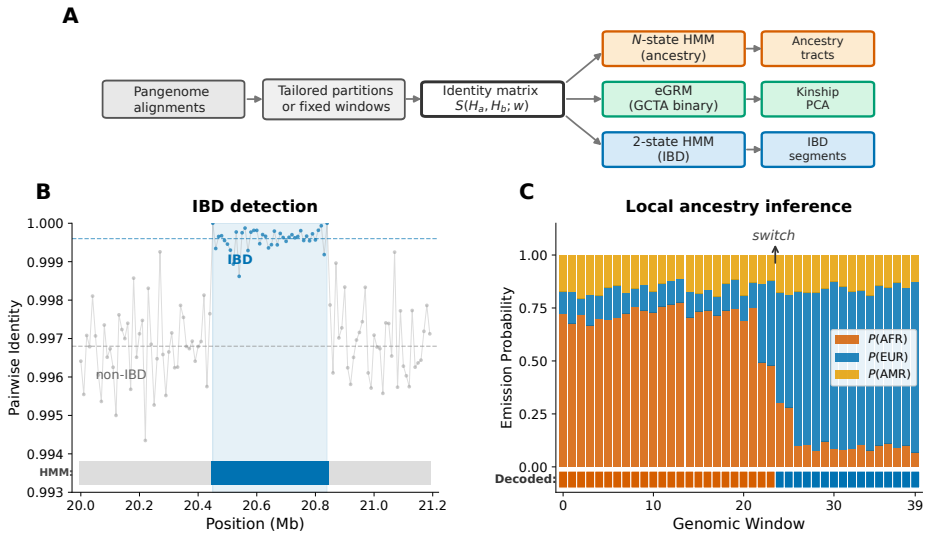


Two answers from the same HMM

Forward-backward	**Viterbi**
marginal posterior *per window*: $P(s_t \mid \mathrm{x})$	most likely *whole-sequence path*: $\arg\max_{s_{1:T}} P(s_{1:T} \mid \mathrm{x})$
smooth, real-valued, integrates over all paths	a single hard call per window, picked jointly
answers "how sure am I about this window?"	answers "what is the single best painting?"
one number per (window, state)	one ancestry label per window

They can disagree. Marginal posterior ≈ 0.5 at a window where Viterbi commits hard. Posterior ≈ 1 where Viterbi was globally over-ruled. Neither is the truth — both are the model's internal belief. The tutorial Part 3 makes this visual on `CHIM_03#1`.

The impop_k pipeline



HMM as a flexible inference framework

Why this skeleton works for both IBD and ancestry:

- Transitions $\equiv$ recombination
- Same architecture: 2 states (IBD), K states (ancestry)
- Emissions are interchangeable

Possible extensions and future directions:

- Improve *graph-aware emissions* on bubble-aware partitions
- *ARG-derived statistics* as alternative observations (genealogical nearest neighbours, coalescent times)
- Joint inference of *IBD and ancestry* on the same identity matrix

Tutorial overview

All four parts on `chr12` long (q) arm — clear of the centromere and telomeres.

1. **IBS** — find structure in the identity matrix (`chr12:40-130 Mb`, CEPH 1463)
2. **IBD** — recover sibling segments (`chr12:40-130 Mb`, CEPH 1463)
3. **Ancestry** — paint admixed haplotypes vs. ground truth (`chr12:50-120 Mb`, 5 chimeras + AFR/EUR panel)
4. **Pedigree painting** — paint grandchildren against their 4 grandparents (`chr12:40-130 Mb`, CEPH 1463, K=4 panel)